Beta-Divergence-Based RPs for Time Series Analysis: Properties and Applications

***Abstract*—Recurrence Plots (RPs) constitute a great tool for revealing structural patterns in time series by measuring proximity in a reconstructed phase space. Traditionally, these distances rely on standard geometric norms such as Euclidean metrics, which assume symmetry and quadratic weighting. In this work, we propose the use of β-divergences as a flexible alternative dissimilarity measure for RP generation. β-divergences form a parametric family that continuously interpolates between well-known divergences.**

Their geometric properties introduce structural behaviours that cannot be replicated by symmetric norms. We analyse how these properties propagate to the geometry of RPs, leading to directional sensitivity, selective emphasis across amplitude ranges, and robustness to outliers. Based on these observations, we derive design guidelines for the construction of β-RP representations, including embedding selection and thresholding rules that are consistent with the divergence scaling laws. This methodological foundation establishes β-RPs as a versatile extension of recurrence analysis for time series.

***Index Terms*—Time Series Analysis, Recurrence Analysis, Recurrence Quantification Analysis, Beta Divergence**

I. Introduction

Recurrence Plots (RPs) are a widely used tool for the analysis and visualization of time series. Since their introduction, they have been successfully applied in many domains, including biomedical signal analysis such as ECG [1] and EEG [2], financial time series [3], and audio and speech processing [4], [5]. Their ability to reveal dynamical structures, transitions, and repetitive patterns has made them a standard exploratory tool in time-series analysis.

Building upon RPs, Recurrence Quantification Analysis (RQA) provides a framework to objectively quantify the structures observed in RPs through a set of descriptive measures [6], [7]. Rather than relying solely on visual inspection, RQA characterizes the temporal organization and complexity of the underlying dynamics by analyzing the distribution and arrangement of recurrence structures, such as diagonal, vertical, and horizontal line patterns.

Beyond standard RQA, the RP framework has been extended to the analysis of complex systems. Recurrence networks interpret the recurrence matrix as a complex network and employ network-theoretic measures to characterize system dynamics [8], [9]. In contrast, recurrence microstate analysis relies on entropy-based descriptors to quantify dynamical complexity [10]. In parallel, recurrence analysis has increasingly been integrated with artificial intelligence techniques for time-series processing. In conventional machine learning pipelines, scalar RQA measures are used as input features for classifiers such as Support Vector Machines or Random Forests [11]. More recently, deep learning approaches treat RPs as image representations, enabling the application of Convolutional Neural Networks and other computer vision models for time-series classification [12], [13].

Despite the widespread success of standard RPs and RQA [6], [14], the reliance on traditional metric distances—primarily the Euclidean or ℓ_p norms—imposes inherent limitations. These symmetric metrics treat all deviations equally, making them highly sensitive to large amplitude variations, extreme outliers, and non-stationary noise. Such artifacts can disproportionately inflate the unthresholded distance matrix and obscure subtle, localized dynamical transitions. Furthermore, standard norms lack the flexibility to adapt to systems exhibiting asymmetric relationships or complex noise profiles. Consequently, we propose the need for more adaptable dissimilarity measures that can be *tailored* to the specific statistical and structural properties of the time series under investigation.

In this work, we propose the use of the *β-divergence* as an alternative for computing the recurrences. The β-divergence generalizes dissimilarity measures, including the Kullback–Leibler divergence [15], the Itakura–Saito divergence [16], and the Squared Euclidean distance [17] as special cases. By replacing the standard distance in the RP construction with the β-divergence, we obtain a family of RPs with tunable sensitivity to amplitude, outliers, and asymmetry. In this work, we show that this generalization introduces new structural properties in the resulting RPs, hereafter referred to as β-RPs, which are analyzed both theoretically and through multiple application domains.

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- Proposal of β-RPs: The introduction of the β-divergence as a flexible, parameterized alternative to classical norms for generating RPs.
- Analysis of Geometric Properties: A theoretical examination of the properties of the β-divergence, and how these properties translate into structural behaviors within β-RPs.
- Novel RQA Metrics: The development of specific Recurrence Quantification Analysis measures tailored to capture the unique scaling and directional dynamics of β-RPs.

- Parameter Selection Criteria: The derivation of principled, guidelines for β-RP construction.

The remainder of this paper is organized as follows. Section II establishes the theoretical background, reviewing classical RPs and formally introducing the β-divergence. Section III analyzes the geometric properties of the β-divergence and their structural implications for the generated β-RPs. Section IV introduces novel RQA measures tailored to capture the unique scaling and directional dynamics of these plots. Section VI derives principled guidelines for parameter selection, . Section VII validates the proposed methodology through three case studies: a visual analysis of the theoretical properties using hydrological data, a quantitative evaluation of the novel RQAs in urban audio signals, and the integration of these metrics into a machine learning classification pipeline. Finally, Section VIII discusses the limitations of the approach, while Section IX concludes the paper and outlines potential extensions.

II. Background

A. RPs

RPs are typically constructed by reconstructing the phase space of a time series using delay embedding [18] with dimension m and delay τ, yielding state vectors

$$\mathbf{x}_i = \big(x_i, x_{i+\tau}, \ldots, x_{i+(m-1)\tau}\big). \tag{1}$$

The unthresholded RP is then defined as the pairwise dissimilarity matrix

$$R(i,j) = d(\mathbf{x}_i, \mathbf{x}_j), \tag{2}$$

where $d(\cdot,\cdot)$ is typically the Euclidean distance or an ℓ_p norm. The standard (binary) RP is obtained by thresholding $R(i,j)$ with a fixed radius ε,

$$\mathrm{RP}(i,j) = \mathbb{I}\big(R(i,j) \le \varepsilon\big), \tag{3}$$

leading to the classical black-and-white representation introduced in [14]. An illustrative example of the unthresholded and thresholded RP construction is shown in Fig. 1.

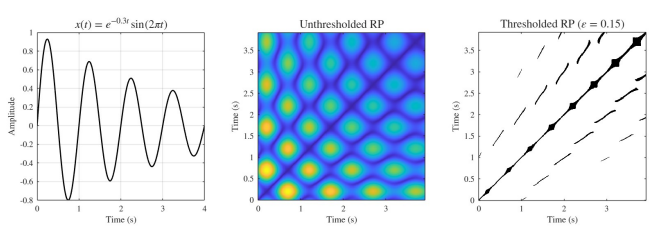

Fig. 1. Example of RP construction for the signal $x(t) = e^{-0.3t}\sin(2\pi t)$. Left: original time series. Middle: unthresholded RP given by the pairwise Euclidean distances between embedded state vectors constructed with embedding dimension $m = 5$ and time delay $\tau = 2$. Right: thresholded RP obtained with threshold $\varepsilon = 0.15$.

B. The β-divergence

In this study, we explore the use of β-divergences as an alternative to classical dissimilarity measures. These divergences form a parametric family governed by a scalar parameter β, enabling a continuous adjustment of the measure according to the desired sensitivity and behavior.

The β-divergence acts as a continuous interpolation between several well-established divergences [19]. Specifically, it encompasses the Itakura–Saito divergence ($\beta = 0$), the Kullback–Leibler divergence ($\beta = 1$), and the Squared Euclidean distance ($\beta = 2$).

Mathematically, β-divergences belong to the class of Bregman divergences [20], [21]. They are defined for scalars $x, y \in \mathbb{R}_+ \setminus \{0\}$ and $\beta \in \mathbb{R}$ as follows:

$$d_\beta(x,y) = \begin{cases} \frac{\left(x^\beta + (\beta-1)y^\beta - \beta x y^{\beta-1}\right)}{\beta(\beta-1)}, & \beta \in \mathbb{R} \setminus \{0,1\}, \\ x \ln \frac{x}{y} - x + y, & \beta = 1, \\ \frac{x}{y} - \ln \frac{x}{y} - 1, & \beta = 0. \end{cases} \tag{4}$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}_+^m$, the β-divergence is applied component-wise and summed across dimensions, yielding the form

$$d_\beta(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{m} d_\beta(x_k, y_k). \tag{5}$$

III. Properties of the β-Divergence and Implications for β-RPs

Unlike classical norms, which are symmetric and homogeneous of degree one, β-divergences possess two distinctive algebraic features: asymmetry and β-order homogeneity. These properties fundamentally shape how dissimilarities are measured across amplitude ranges and under argument permutation, and therefore directly influence the structure of the resulting RPs. In the following, we formalize these two properties and later analyze how they propagate to the geometry and interpretability of β-based RPs.

A. Geometric Properties

Lemma 1 (Homogeneity of Degree β)**.** *The β-divergence is homogeneous of degree β. For all $c > 0$ and $x, y \in \mathbb{R}_+^m$:*

$$d_\beta(cx \mid cy) = c^\beta \, d_\beta(x \mid y) \tag{6}$$

Proof. Although this property is stated in [19], [22], the proof is straightforward and follows from direct substitution.

Lemma 2 (Asymmetry)**.** *For any scalars $x, y \in \mathbb{R}_+$ with $x > y$, the β-divergence satisfies*

$$\begin{cases} d_\beta(x \mid y) > d_\beta(y \mid x), & \beta < 2, \\ d_\beta(x \mid y) = d_\beta(y \mid x), & \beta = 2, \\ d_\beta(x \mid y) < d_\beta(y \mid x), & \beta > 2. \end{cases} \tag{7}$$

Moreover, for fixed $\beta \neq 2$, the absolute asymmetry

$$\Delta_\beta(x,y) := \big| d_\beta(x \mid y) - d_\beta(y \mid x) \big| \tag{8}$$

increases with $(x - y)$ and vanishes at $x = y$.

Finally, for fixed $x > y$, the ratio

$$\mathcal{R}(\beta) := \frac{d_\beta(y \mid x)}{d_\beta(x \mid y)} \tag{9}$$

is strictly increasing in $\beta \in \mathbb{R}$ (with $\mathcal{R}(2) = 1$).

Proof See Appendix.

B. Implications of Properties for β-RPs

The properties defined in the previous section directly influence the generated RPs. In this subsection, we analyze the specific novelties contributed by each property highlighting the features they introduce relative to classical norms.

1) Impact of Homogeneity: As a direct consequence of the homogeneity property (Lemma 1), the β-divergence applies a non-uniform weighting to discrepancies across the signal's range. The parameter β modulates the metric's "focus":

- Lower values of β increase sensitivity to relative variations among *low-valued* components.
- Higher values of β emphasize discrepancies among *large-valued* components.

This flexibility is particularly advantageous when analyzing data that departs from Gaussianity or contains skewed outliers. In such cases, the fixed quadratic weighting of the Euclidean case ($\beta = 2$) may distort the underlying structure by disproportionately emphasizing extreme observations. By selecting an appropriate β, the RP generation becomes robust to these artifacts, allowing the analysis to target specific amplitude ranges.

We illustrate this behavior in Figures 2 and 3, which demonstrate how varying β shifts the recurrence structure's sensitivity between "normal" dynamics and extreme events.

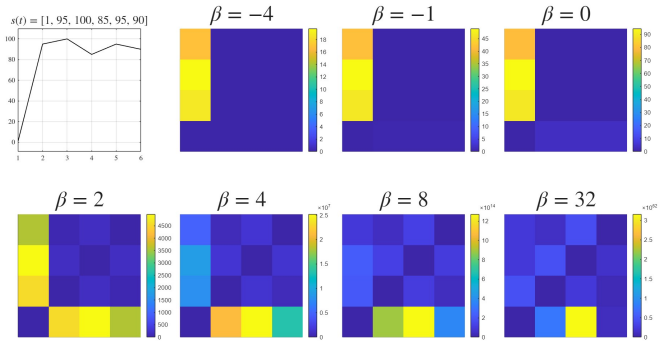

Fig. 2. Illustration of the robustness of β-divergence-based RPs to outliers. The top left panel shows a time series containing a clear outlier at $t = 1$. For $\beta = -4$, the outlier *collapses* the entire recurrence pattern, as the RP mainly captures the discrepancies between the normal values and the outlier. As β increases, the influence of this extreme observation gradually decreases, and the RP progressively focuses on the relationships among the *normal* samples. For all the RPs embedding dimension was $m = 3$ and delay $\tau = 1$.

2) Impact of Asymmetry: The asymmetry described in Lemma 2 manifests in the RP texture by distinguishing between the order of the arguments in the divergence.

To illustrate this, Figure 4 displays the RPs of a unit pulse signal. While the Euclidean case ($\beta = 2$) produces perfectly symmetric tiles, deviating from it introduces a progressive asymmetry.

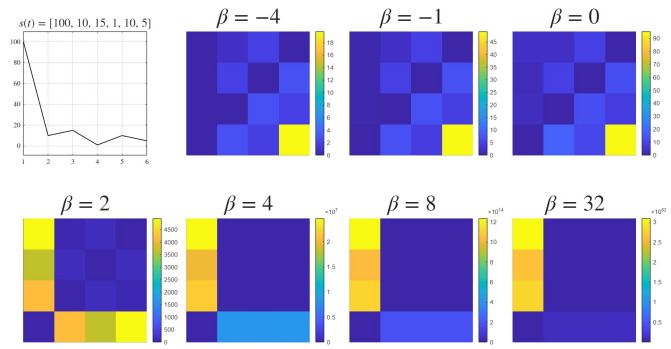

Fig. 3. Complementary case where the outlier exceeds the typical data amplitude. In this scenario, for $\beta = -4$, the recurrence pattern remains less affected, while increasing β progressively amplifies the contrast around the outlier. For all the RPs embedding dimension was $m = 3$ and time delay $\tau = 1$.

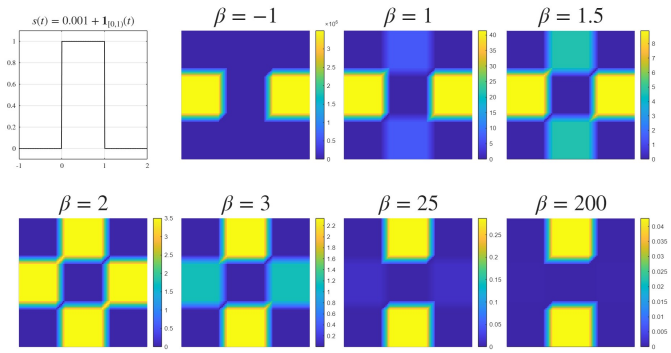

Fig. 4. Effect of asymmetry on a unit pulse. The symmetry of the recurrence blocks at $\beta = 2$ dissolves into asymmetric patterns as $|\beta - 2|$ increases. For all the RPs embedding dimension was $m = 7$ and time delay $\tau = 3$.

This sensitivity is further highlighted when analyzing time-reversed signals. While symmetric norms impose structural symmetry on the recurrence matrix, potentially obscuring directional nuances depending on the signal's properties, β-divergences can encode directionality through their asymmetry. As shown in Figure 5, β-divergences sometimes resolve ambiguity that may arise in the Euclidean case.

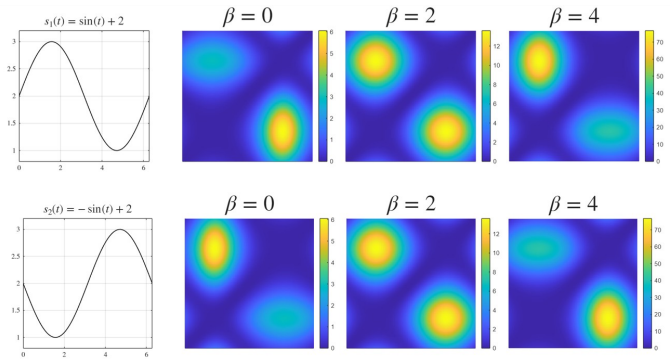

Fig. 5. Directional sensitivity test. The central column ($\beta = 2$) shows identical plots, whereas the outer columns exhibit complementary asymmetric structures that clearly distinguish temporal direction. For all the RPs embedding dimension was $m = 7$ and time delay $\tau = 2$.

IV. Novel RQA Metrics for β-RPs

Standard RQAs were originally developed to characterize the structures found in symmetric RPs based on classical norms [6]. Because β-RPs are generally asymmetric and sensitive to signal scaling, traditional RQA measures may overlook some informative dynamical features captured by this divergence family. The following subsections propose a set of RQA measures specifically designed to quantify the behaviors of β-RPs. These measures are intended to reflect the two core geometric properties analyzed in Section III.

V. Novel RQA Metrics for β-RPs

Before introducing measures tailored to β-RPs, it is useful to review the established RQA metrics. Standard RQA measures (the key ones summarized in Table I) were originally developed to characterize the structures found in symmetric RPs based on classical norms [6].

TABLE I
Standard RQA Metrics

Metric	Formula	Description
Recurrence Rate (RR)	$RR = \frac{1}{N^2}\sum_{i,j=1}^{N} R_{i,j}$	Density of recurrence points in the RP. Represents the probability that a specific state recurs.
Determinism (DET)	$DET = \frac{\sum_{l=l_{min}}^{N} l\,P(l)}{\sum_{i,j=1}^{N} R_{i,j}}$	Ratio of recurrence points forming diagonal structures. Indicates the predictability of the system.
Average Diagonal (L)	$L = \frac{\sum_{l=l_{min}}^{N} l\,P(l)}{\sum_{l=l_{min}}^{N} P(l)}$	Average length of diagonal lines. Related to the divergence of system trajectories.
Entropy (L_{entr})	$L_{entr} = -\sum_{l=l_{min}}^{N} p(l)\ln p(l)$	Shannon entropy of the probability distribution of diagonal line lengths.
Laminarity (LAM)	$LAM = \frac{\sum_{v=v_{min}}^{N} v\,P(v)}{\sum_{i,j=1}^{N} R_{i,j}}$	Ratio of recurrence points forming vertical structures. Represents the occurrence of laminar states.
Trapping Time (TT)	$TT = \frac{\sum_{v=v_{min}}^{N} v\,P(v)}{\sum_{v=v_{min}}^{N} P(v)}$	Average length of vertical lines. Indicates how long the system remains trapped in a state.

Note: N is the number of states, $R_{i,j}$ is the recurrence matrix, $P(l)$ and $P(v)$ are the frequency distributions of diagonal lines of length l and vertical lines of length v, respectively. l_{min} and v_{min} are the minimum line lengths. $p(l) = P(l)/\sum_{l=l_{min}}^{N} P(l)$ is the probability of finding a diagonal line of exactly length l.

However, because β-RPs are generally asymmetric and uniquely sensitive to signal scaling, these traditional measures may overlook informative dynamical features captured by this divergence family. To address this limitation, the following subsections propose a set of novel RQA measures specifically designed to quantify the distinct behaviors of β-RPs. These measures are intended to reflect the two core geometric properties analyzed in Section III.

A. RQAs for Quantifying Homogeneity

As established by the homogeneity property (Lemma 1), the parameter β acts as a selective filter across the signal's dynamic range. To characterize how this sensitivity shift affects the recurrence structure, we propose the following scaling-aware measures.

1) Laminarity Scaling Index (LSI): Laminarity (LAM) quantifies the prevalence of vertical and horizontal line structures in the RP, typically associated with persistence of the system in a given state. In β-RPs, these structures may be enhanced or suppressed depending on how the chosen value of β weights amplitude differences across the signal.

Let $a_i = \|\mathbf{x}_i\|_2$ be the Euclidean norm, let $\mathbf{x}_i \in \mathbb{R}^m$ denote the embedded state at time i, and let q_p denote the p-quantile of the sample of amplitudes $\{a_i\}_{i=1}^{N}$.

We define the subsets of low- and high-amplitude states as

$$\mathcal{X}_{\text{low}}(p) = \{\mathbf{x}_i : a_i \le q_p\}, \tag{10}$$

$$\mathcal{X}_{\text{high}}(p) = \{\mathbf{x}_i : a_i \ge q_{1-p}\}. \tag{11}$$

These subsets induce vertical and horizontal regions in the RP. The vertical laminarity conditioned on high- and low-amplitude states is computed by restricting vertical line structures to recurrences whose reference state belongs to the corresponding subset, yielding

$$LSI^{(v)}(p) = \frac{LAM_v(\mathcal{X}_{\text{high}}(p))}{LAM_v(\mathcal{X}_{\text{low}}(p))}. \tag{12}$$

Analogously, restricting horizontal structures with respect to the second state gives

$$LSI^{(h)}(p) = \frac{LAM_h(\mathcal{X}_{\text{high}}(p))}{LAM_h(\mathcal{X}_{\text{low}}(p))}. \tag{13}$$

Finally, the Laminarity Scaling Index is defined as

$$LSI(p) = \frac{1}{2}\left(LSI^{(v)}(p) + LSI^{(h)}(p)\right). \tag{14}$$

This RQA characterizes whether laminar behavior is more pronounced around high-amplitude events ($LSI > 1$) or concentrated in low-amplitude regimes ($LSI < 1$), thereby providing a β-dependent assessment of amplitude-specific stability. As established by the homogeneity property (Lemma 1) small values of β are expected to emphasize differences among low-amplitude components while compressing differences at high amplitudes, which is expected to decrease LAM within low-amplitude regions and simultaneously enhance laminar structures associated with high-amplitude events, leading to greater LSI value. Conversely, large values of β amplify differences at high amplitudes and compress low-amplitude differences, favoring laminarity in low-amplitude, and therefore yielding lower LSI value. In Figure 6, this metric is illustrated for the first signal of the example shown in Figure 5.

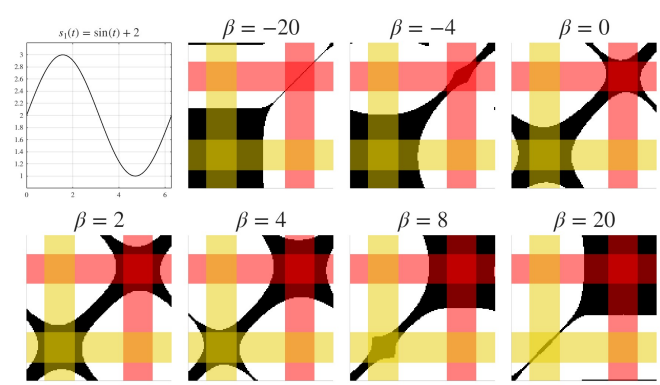

Fig. 6. Illustration of the computation of LSI for different values of β. The red and yellow grids represent vectors with norms below and above, respectively, the selected percentile ($p = 0.2$). As β increases, laminarity becomes greater in the red subset (lower values) and smaller in the subset associated with higher-amplitude vectors. The embedding dimension is $m = 7$, the time delay is $\tau = 2$, and ε is chosen in each RP to guarantee a recurrence rate of 30% of the total points.

2) Amplitude-Weighted Recurrence Rate (RR_{amp}): To measure the *focus* of a β-RP relative to the signal magnitude, we evaluate the mean local amplitude associated with the recurrence points. Let $a_i = \|\mathbf{x}_i\|_2$ be the Euclidean norm. We define:

$$RR_{\text{amp}}(\beta) = \frac{\sum_{i \neq j} R_{i,j}(\beta) \cdot \frac{1}{2\sqrt{m}}(a_i + a_j)}{\sum_{i \neq j} R_{i,j}(\beta)}. \tag{15}$$

In contrast to LSI, which evaluates how laminar structures redistribute across amplitude regimes, RR_{amp} captures the average amplitude level at which recurrences occur. Its evolution with β therefore provides a complementary indicator of how the metric shifts the dominant recurrence region along the signal magnitude, without relying on laminarity structure.

Nevertheless, a behavior consistent with LSI is expected. Smaller values of β enhance dissimilarities among low-amplitude components while compressing differences at high amplitudes, which shifts recurrences toward higher-amplitude states and consequently increases RR_{amp}. Conversely, larger values of β accentuate differences at high amplitudes, promoting recurrences in lower-amplitude regions and therefore decreasing RR_{amp}.

B. RQAs for Quantifying Asymmetry

Unlike classical RPs, where the recurrence matrix is symmetric by definition ($R_{i,j} = R_{j,i}$), β-RPs inherently encode directional information due to Lemma 2. To characterize this, we propose a set of directional RQA measures that treat the upper ($i < j$) and lower ($i > j$) triangles of the matrix independently.

1) Recurrence Rate Asymmetry: The Recurrence Rate Asymmetry (RR_{asym}) is defined as the normalized difference between the recurrence densities of the two triangles:

$$RR_{\text{asym}} = \frac{RR_{up} - RR_{lo}}{RR_{up} + RR_{lo}}, \tag{16}$$

where RR_{up} and RR_{lo} are the recurrence rates calculated for the upper and lower triangles, respectively. A non-zero RR_{asym} indicates a bias in the proximity of states depending on the temporal order of the comparison. It is important to note that this formulation is generalizable: the Recurrence Rate (RR) can be substituted by other conventional RQA measure, such as Laminarity (LAM) or Determinism (DET), to analyze specific aspects of structural asymmetry.

For $\beta < 2$, the RP will exhibit higher density (higher RR) in regions where the first argument of the divergence is small. Conversely, for $\beta > 2$, higher recurrence density is observed in regions where the second argument of the divergence is small. Consequently, this RQAs allow for the quantification of asymmetric dynamical behaviors that are invisible to standard geometric norms.

2) Unreciprocated Recurrence Rate (URR): While RR_{asym} captures global differences in recurrence density between the upper and lower triangles, it does not strictly quantify the violation of geometric symmetry at the local level. It is possible for RR_{up} and RR_{lo} to be equal even if the recurrence structures are spatially distinct. In Figure 7, we illustrate an instance of this behavior using the same example shown in Fig. 4.

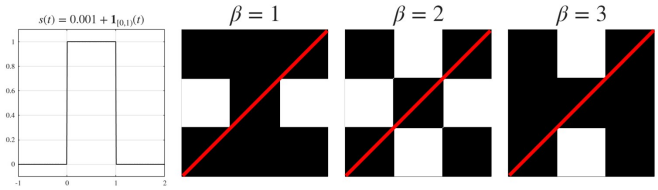

Fig. 7. Illustration of thresholded RPs. The red line separates the lower and upper triangles. For $\beta = 1$ and $\beta = 3$, the recurrence rate is identical in both triangles, however recurrences do not occur on symmetric tiles ($i \neq j$). For $\beta = 2$, the two triangles are symmetric due to the symmetry of the squared Euclidean distance.

To quantify the explicit pixel-wise violation of symmetry, we define the *Unreciprocated Recurrence Rate* (URR). The URR represents the proportion of recurrence points that lack a symmetric counterpart. By iterating over the lower triangle ($i > j$), the measure can be compactly defined as:

$$URR = \frac{2\sum_{i>j} |R_{i,j} - R_{j,i}|}{\sum_{i>j}(R_{i,j} + R_{j,i})}. \tag{17}$$

The URR provides information complementary to RR_{asym}. While both metrics vanish for symmetric RPs ($\beta = 2$), they diverge in their characterization of asymmetry. RR_{asym} is a signed measure that identifies the directionality of the asymmetry, whereas URR is an unsigned measure quantifying the total volume of unreciprocated points. While a high $|RR_{\text{asym}}|$ mathematically necessitates a high URR, the converse is not true: a system may exhibit high structural non-reciprocity ($high\ URR$) without a net density bias ($low\ RR_{\text{asym}}$). Thus, both indices are required to fully characterize the breakdown of symmetry in β-RPs.

VI. Parameter Selection for β-RPs

The construction of $\beta - RPs$ requires the determination of four parameters: the time delay τ, the embedding dimension m, β, and, specifically for thresholded RPs, the cutoff distance ε. While some standard heuristics exist for Euclidean RPs, the introduction of the β-divergence requires a re-evaluation of these selection criteria.

A. Time Delay (τ)

The time delay τ is selected to reduce redundancy while preserving the dynamical structure of the time series. Since this dependency is an intrinsic property of the time series and not of the proximity measure used in the RP, standard criteria can be employed. In this work, τ is determined using the first minimum of the Average Mutual Information (AMI) [23]. To avoid unrealistically large delays in finite signals, the search is restricted to a maximum value given by $\lceil 0.025\, N \rceil$, where N denotes the length of the time series. Nevertheless, any established method for time-delay selection in recurrence analysis would be equally valid.

B. Embedding Dimension (m) via β-FNN

The embedding dimension m must be sufficiently large to avoid false crossings in the reconstructed phase space. The classical False Nearest Neighbors (FNN) method [24] selects the minimal m for which neighbor distances stabilize as the dimension increases. However, since standard FNN relies on the Euclidean norm, it may yield suboptimal dimensions for β-RPs, where neighborhood relations depend on the geometry induced by the β-divergence. Points that are neighbors under the Euclidean metric may not be neighbors under d_β, and vice versa.

Let $\mathbf{x}_i^{(m)}$ denote the state vector at time i in dimension m, and $\mathbf{x}_{n(i)}^{(m)}$ its nearest neighbor according to d_β. A neighbor is considered *false* if the divergence increases disproportionately when moving to dimension $m+1$, quantified by

$$R_i(m,\beta) = \frac{d_\beta\big(\mathbf{x}_i^{(m+1)}, \mathbf{x}_{n(i)}^{(m+1)}\big)}{d_\beta\big(\mathbf{x}_i^{(m)}, \mathbf{x}_{n(i)}^{(m)}\big)}. \tag{18}$$

By the homogeneity of d_β (Lemma 1), this ratio is scale-invariant. We define a data-driven tolerance as the $(1-\alpha)$-quantile of $\{R_i(m,\beta)\}_{i=1}^N$,

$$R_{\mathrm{tol}}(m,\beta) = Q_{1-\alpha}\big(\{R_i(m,\beta)\}_{i=1}^N\big), \tag{19}$$

and classify a neighbor as false if $R_i(m,\beta) > R_{\mathrm{tol}}(m,\beta)$. In this work, we use $\alpha = 0.05$ (95th percentile).

The embedding dimension is selected as the smallest m for which the fraction of false neighbors vanishes or reaches a stable plateau. The plateau is detected using an elbow criterion: we select the first m such that

$$\frac{\mathrm{FNN}(m-1) - \mathrm{FNN}(m)}{\mathrm{FNN}(m-1)} < 0.01, \tag{20}$$

and remains below this level for subsequent dimensions.

When comparing β-RPs across different β values, a common phase-space reconstruction is required to ensure comparability. Using different embedding dimensions would confound geometric effects with reconstruction changes. Let $\mathcal{B} \subset \mathbb{R}$ denote the finite set of β values considered in the analysis. After estimating m_β for each $\beta \in \mathcal{B}$, we adopt a shared embedding

$$m = \max_{\beta \in \mathcal{B}} m_\beta, \tag{21}$$

ensuring that differences between β-RPs arise solely from the divergence parameter.

C. The Recurrence Threshold ε

For thresholded RPs, the selection of ε is the parameter most critically affected by the choice of β.

Homogeneity of degree β (Lemma 1) implies that the divergence changes systematically with β. As β varies, the overall magnitude of the pairwise distances $d_\beta(\mathbf{x}_i, \mathbf{x}_j)$ changes, causing the distance distribution to shift and rescale even when the dataset remains fixed.

Consequently, using a fixed absolute value for ε across different β values is methodologically unsound, as it would result in vastly different recurrence rates.

To ensure consistency with the metric's geometry, the threshold selection must satisfy the Homogeneity property. We define the threshold as a function $\varepsilon_\beta(X)$ of the embedded trajectory X, imposing the scaling constraint:

$$\varepsilon_\beta(cX) = c^\beta\, \varepsilon_\beta(X), \quad \forall c > 0. \tag{22}$$

Any thresholding strategy respecting this condition is admissible. We highlight three approaches:

- Quantile-based (Fixed RR): Selecting ε_β as a fixed quantile of the pairwise distance distribution. This enforces a constant recurrence rate across different β values and is scale-consistent by construction.
- Dispersion-based (MAD): Defining $\varepsilon_\beta = k \cdot \mathrm{MAD}(D_\beta)$, where MAD denotes the median absolute deviation of the distances. This approach satisfies Eq. (22) and offers superior robustness against outliers compared to mean-based scaling.
- Nearest-neighbour radius (NNR): Deriving the threshold from local neighborhood scales to adapt to the manifold density. For instance, using the median of the k-th nearest neighbor distances:

$$\varepsilon_\beta = \lambda \cdot \mathrm{median}_i\left(r_k^{(i)}\right), \tag{23}$$

  where $r_k^{(i)}$ is the distance from state $\vec{x}_i$ to its k-th nearest neighbor under d_β.

In this work we will mainly use the Quantile-based approach. Nevertheless, any other construction that respects the homogeneity condition in Eq. (22) could be in principle valid, and the optimal choice depends on the specific robustness needs or application constraints.

D. Selection of β

The selection of the β parameter is not merely a tuning exercise, but is inherently task-dependent. Rather than yielding a universally optimal value, any selection criterion can only be optimal with respect to a specific objective or structural property of interest. Consequently, our goal is not to identify a single optimal β for all applications, but to offer selection strategies aligned with different analytical needs.

These strategies address different objectives. The first maximizes contrast and effective use of the metric range, the second promotes amplitude-balanced recurrence structures, and the third selects β based on a task-defined criterion.

1) Maximization of Recurrence Contrast and Entropy: An effective β should maximize the discriminatory power of the induced dissimilarity measure. If β is poorly chosen, the distribution of distances $d_\beta(\mathbf{x}_i, \mathbf{x}_j)$ may become either too narrow or overly dispersed, leading to RPs that are excessively sparse or saturated, and therefore structurally uninformative.

We define the *distance distribution entropy* $H(d_\beta)$ as

$$H(d_\beta) = -\sum_{k=1}^{K} p(z_k)\, \ln p(z_k), \tag{24}$$

where $z_k \in [a_k, a_{k+1})$ denotes the k-th bin of the empirical distribution of pairwise β-divergence distances $d_\beta(\mathbf{x}_i, \mathbf{x}_j)$, and K is a fixed number of bins (independent of β). A value of

β that maximizes $H(d_\beta)$ encourages a non-degenerate use of the metric range, avoiding overly concentrated or saturated distance distributions.

However, the optimal entropy value is not known *a priori* and must be estimated numerically. While this approach often yields rich recurrence structures (frequently reflected in higher Shannon entropy of diagonal line lengths, L_{ENTR}), it is inherently exploratory and computationally demanding.

2) Amplitude-Consistent Recurrence Structure: A natural reference configuration for β is one in which the recurrence structure does not exhibit systematic bias towards either low- or high-amplitude regimes. In terms of the homogeneity-aware measures introduced previously, this corresponds to a balanced scaling regime where laminar behavior is evenly represented across amplitudes and the average recurrence amplitude reflects the intrinsic signal scale.

Two alternative selection criteria can be considered:

$$LSI(\beta) \approx 1, \qquad RR_{\text{amp}}(\beta) \approx \bar{a}, \tag{25}$$

where $\bar{a} = \frac{1}{N}\sum_{i=1}^{N} a_i$ denotes the mean amplitude of the embedded state vectors, with $a_i = \|\mathbf{x}_i\|_2$. These correspond to different strategies: the first enforces amplitude-neutral laminarity, whereas the second centers recurrence around the natural signal scale. They are presented together due to their structural similarity, but are not jointly optimized.

Both criteria are attractive for three main reasons. First, they are interpretable, as their target values are known *a priori*. Second, they are computationally efficient, requiring only evaluation of RQA measures over a one-dimensional grid of β. Moreover, the homogeneity-induced monotonic behavior of LSI provides directional guidance for the search: $LSI > 1$ suggests increasing β, while $LSI < 1$ indicates decreasing β. The same directional behavior is observed for RR_{amp}, enabling guided rather than exhaustive search. Finally, these criteria promote structurally balanced RPs that are less sensitive to amplitude distortions induced by homogeneity.

It is important to mention that the sensitivity of the LSI is governed by the chosen percentile p, which defines the selectivity of the amplitude regimes. The parameter is naturally bounded by $p \leq 0.5$, where the maximum corresponds to a partition based on the median state vector norm. Reducing p restricts the evaluation to the distribution tails, emphasizing stability contrasts between extreme high- and low-value events. Consequently, the selection of p is application-dependent, allowing the index to be tuned for either global regime characterization ($p \approx 0.5$) or rare-event stability analysis ($p \ll 0.5$) according to the specific data distribution.

3) Task-Driven Selection of β: In many practical settings, the choice of β is not guided by structural balance but by the specific objective of the analysis. Depending on the application, β may be selected to optimize any relevant performance criterion derived from recurrence representations.

Typical strategies include maximizing or minimizing a chosen RQA statistic, targeting a predefined value of a structural measure, or directly optimizing downstream task performance. For instance, when recurrence features or β-RPs are used within a learning pipeline, β can be chosen to maximize classification or prediction metrics such as accuracy or F1-score:

$$\beta^\star = \arg\max_{\beta}; \mathcal{J}(\beta), \tag{26}$$

where $\mathcal{J}(\beta)$ denotes a task-dependent objective, e.g., an RQA measure, a target structural criterion, or an external performance metric.

VII. Case Studies

In this section, we validate the methodological framework proposed above through distinct case studies. We explore two distinct application domains. Case I uses hydrological time series to visually demonstrate how the asymmetry and homogeneity of the β-divergence capture specific amplitude regimes and directional behaviors. Case II analyzes urban audio signals to address the divergence's strict positivity constraint and quantitatively validate our proposed β-RQA measures.

A. Case I: Hydrologic Dynamics

Note: I have yet to decide whether to use thresholded or unthresholded RPs for this section. I believe unthresholded RPs are more informative, but thresholded ones are more standard.

In this section, we apply distinct β parameters to three different hydrological datasets to demonstrate specific properties of the beta divergences claimed in Section III-B. The time delay τ and the embedding dimension m are determined using the methods described in Subsections VI-A and VI-B, respectively.

1) River Discharge: Daily river discharge data were obtained from the Lobith station, which monitors the Rhine River at the point where it enters the Netherlands. The dataset is provided by Rijkswaterstaat through its public Waterinfo platform [25].

Figure 8 shows a 9-month discharge record together with the corresponding β-divergence-based RPs obtained for different values of β.

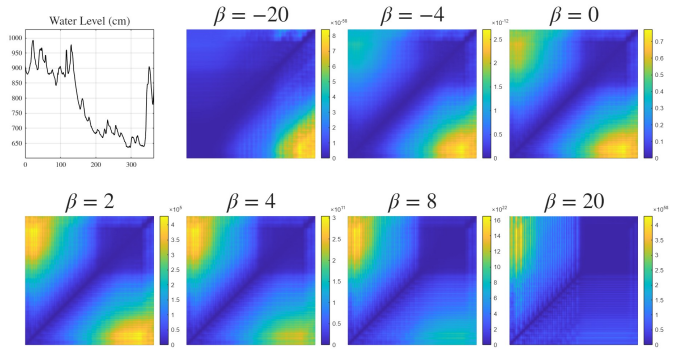

Fig. 8. Daily Rhine River discharge at Lobith over a 9-month period in 2018, including a drought period, and the corresponding β-divergence-based RPs for different values of β. For all the RPs embedding dimension was $m = 14$ and time delay $\tau = 9$.

This example illustrates the asymmetry property discussed in Section III-B2. For smaller values of β, the largest divergence values concentrate in the bottom-right block, corresponding to comparisons between high-discharge and drought

values, where the high values appear as the first argument of the divergence. As β increases, the region of highest divergence shifts to the top-left block, which represents the same type of comparison but with the lower values as the first argument.

The effect of the homogeneity property described in Section III-B1 is also visible. The bottom-left square captures differences among the high discharge values at the beginning of the series, which become more pronounced as β increases. In contrast, the top-right square reflects differences among the low-flow drought values, whose visibility diminishes for larger β.

2) Tidal Dynamics: In this example we analyze water level data from NOAA station 8443970 (Boston, MA) [26]. Figure 9 displays the RPs for $\beta \in \{0, 2, 4\}$ over periods of approximately 2 days (16-18 and 24-26 October 2025), with 1 data point every 6 minutes.

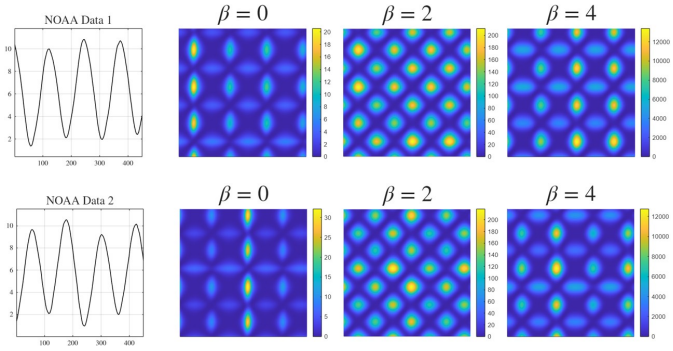

Fig. 9. NOAA tidal data over two different periods and corresponding β-divergence-based RPs for different values of β. For all the RPs embedding dimension was $m = 5$ and time delay $\tau = 3$.

This example illustrates the effect of sensitivity to directionality discussed in Section III-B2. The two signals are sine-like functions: the first one starts and ends near a maximum, while the second one starts and ends near a minimum. For $\beta = 2$, the resulting RPs are nearly identical for both signals, as the Euclidean case mainly highlights where the pointwise differences in the signal are largest, without a clear directional preference.

The effect of the homogeneity property described in Section III-B1 is also visible in these results. Vertical and horizontal line structures in the RPs correspond to laminar states, i.e., time intervals during which the system remains in (or very close to) the same state.

For small $\beta = 0$, the homogeneity of the β-divergence emphasizes low-amplitude components of the signal, effectively emphasizing distances around local minima. As a result, the time indices associated with these minima become differentiated from the rest of the signal, giving rise to vertical and horizontal dotted lines. Conversely, for larger values of β, the emphasis shifts towards high-amplitude components. Accordingly, for $\beta = 4$, similar laminar line structures appear, now aligned with the local maxima of the signals.

3) Precipitation: Precipitation time series are intrinsically zero-inflated and heavy-tailed. The daily precipitation data used in this experiment were obtained from the European Climate Assessment & Dataset (ECA&D) station for Maastricht (Helmstraat), The Netherlands [27]. Figure 10 shows a representative 250-day time window of precipitation together with the corresponding RPs obtained for different values of β.

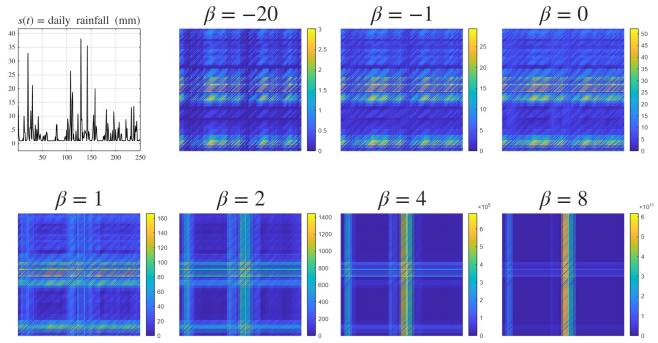

Fig. 10. Daily precipitation time series over a 250-day period and corresponding β-divergence-based RPs for different values of β. For all the RPs embedding dimension was $m = 8$ and time delay $\tau = 2$.

We can see in the example that, as β increases, the influence of large-amplitude deviations becomes amplified. Consequently, for $\beta \geq 2$, the RP emphasizes intense precipitation events.

This directly reflects the theoretical behavior illustrated in Figure 3. Where the β-divergence acts as a tunable mechanism that shifts the RP sensitivity.

B. Case II: Urban Audio Dynamics

In this subsection, we center our evaluation on the analysis of acoustic signals using the UrbanSound8K dataset [28]. The dataset consists of short audio clips of up to 4 seconds in duration, categorized into 10 urban sound classes. The time delay τ and the embedding dimension m are determined using the methods described in Subsections VI-A and VI-B, respectively. The threshold ε is determined by enforcing a recurrence rate of $\mathrm{RR} = 0.1$.

1) Addressing the Strict Positivity Constraint: A fundamental requirement of the β-divergence is that its arguments must be strictly positive, $x, y \in \mathbb{R}_+ \setminus \{0\}$. To map these signals into the valid domain, the most straightforward approach is applying a global constant shift, $x'_i = x_i - \min(x) + \epsilon$. While computationally trivial, this transformation fundamentally alters the relative scale between data points, thereby interfering non-trivially with the β-order homogeneity property (Lemma 1) and distorting the metric's natural amplitude weighting.

To strictly preserve the scaling laws of the β-divergence, we map the audio into the positive domain by computing its local energy envelope via a rolling Root Mean Square (RMS). Because the RMS operator is scale-equivariant, ensuring that an amplitude scaling factor c is perfectly preserved in the envelope, the transformed strictly positive signal maintains the exact homogeneity behaviors theoretically derived in Section III.

2) Quantitative Evaluation via β-RQAs: To quantify how the choice of β interacts with these distinct sound profiles, we employ the scale-aware and directional RQA measures

introduced in Section III-B. Table II summarizes the evolution of these metrics for different values of β across three selected acoustic profiles, the reported values correspond to averages over up to 30 audio files per class, computed from the normalized RMS envelope (600 samples) of each signal.

TABLE II
QUANTITATIVE EVALUATION OF β-RPS ACROSS URBAN AUDIO PROFILES.

Acoustic Profile	β	RR_{amp}	$LSI(p = 0.15)$	RR_{asym}	URR
Gun Shot	0	0.159	1.107	-0.039	0.254
	1	0.135	0.520	-0.022	0.134
	2	0.128	0.287	0.000	0.000
	4	0.124	0.048	-0.001	0.166
Siren	0	0.031	0.808	-0.010	0.183
	1	0.030	0.618	-0.006	0.089
	2	0.029	0.460	0.000	0.000
	4	0.029	0.236	0.018	0.146
Air Conditioner	0	0.095	0.564	0.001	0.119
	1	0.094	0.459	0.000	0.059
	2	0.094	0.376	0.000	0.000
	4	0.094	0.259	-0.002	0.106

As observed in Table II, the empirical values of LSI and RR_{amp} align with the theoretical scaling behaviors derived from the homogeneity property of the β-divergence. Across all three acoustic profiles, increasing the divergence parameter β yields a monotonic decrease in both LSI and RR_{amp}. Notably, this amplitude-weighting effect is most pronounced for the "Gun Shot" profile, which is a sound profile characterized by extreme peaks. For this class, $\beta = 0$ slightly emphasizes the high-amplitude transient ($LSI \approx 1.1$), whereas higher β values rapidly suppress its influence, driving the LSI down to 0.048 at $\beta = 4$. In contrast, the more continuous "Siren" and "Air Conditioner" signals exhibit more stable RR_{amp} and LSI values. Finally, it is important to note that the standard squared Euclidean distance ($\beta = 2$) yields unbalanced structures ($LSI \ll 1$), skewing the recurrence density toward low-amplitude regimes.

Note:

I could add extra information depending on the ideal length for the paper and for this section. Some possible ideas I find interesting are:

- Comparison with Euclidean distance.
- Search for the optimal β according to the proposed criteria.
- Inclusion of signal and RP images to visually explain these RQA values.

C. Case III: Machine learning

This subsection is still in progress. I could cite Filip's work [11], but I don't like referring to a predictor based only on classical RQA measures rather than the ones introduced here. I am currently leaning towards using the Case II: Urban Audio results to construct a simple classifier, as it would be straightforward to implement and would not require much additional space. We can discuss this in the next meeting.

VIII. DISCUSSION

The introduction of the β-divergence into recurrence analysis provides a flexible framework for time series characterization, enabling tunable sensitivity to amplitude variability and structural asymmetry. This generalization, however, also entails theoretical and practical challenges that must be carefully considered. From a computational perspective, the β-RP framework is inherently more demanding than classical RP analysis, since a comprehensive exploration often requires computing distance matrices for multiple β values rather than relying on a single geometric norm.

A further limitation arises from the mathematical requirement of the β-divergence for strictly positive inputs ($x, y > 0$). When signals contain zero or negative values, preprocessing becomes necessary and introduces nontrivial trade-offs. Global shifting, obtained by subtracting the minimum and adding a small constant, is simple but modifies the relative scale of the data; since the β-divergence is homogeneous of degree β, this rescaling directly interacts with its amplitude-weighting behavior. Alternatively, splitting the signal into positive and negative components preserves relative scaling but requires replacing exact zeros with a small regularization constant ε, which may artificially increase local density near zero. In some modalities, scale-equivariant envelope representations such as rolling RMS provide a practical workaround, yet the treatment of raw signals with strong negative components remains application-dependent.

Regarding parameterization, this study employed conventional heuristics for phase-space reconstruction, including average mutual information for delay selection and a modified β-FNN criterion for embedding dimension. While effective, the flexibility of the β-divergence suggests that alternative strategies could further improve geometric consistency, particularly in extreme β regimes. In particular, the interaction between threshold selection rules and the asymmetry and homogeneity properties of the divergence remains a key methodological consideration.

Beyond classical analysis, the β-RP framework shows strong potential for integration into machine learning pipelines. Image-based deep learning architectures have proven effective for recurrence-based representations, and in this context β can be treated as a tunable hyperparameter optimized with respect to downstream objectives such as classification or anomaly detection. This is particularly relevant in biomedical and acoustic settings, where amplitude-dependent or transient dynamics may not be adequately captured by Euclidean-based recurrence measures.

IX. CONCLUSION AND FUTURE WORK

By leveraging the asymmetry and β-order homogeneity of the β-divergence family, the proposed β-RPs provide tunable sensitivity to amplitude regimes, improved robustness to outliers, and the ability to capture directional dynamics. The methodological framework introduced in this work, including parameter selection and β-aware RQA measures, enables an analysis of asymmetric and amplitude-weighted recurrence structures.

Empirical results across hydrological and acoustic signals show that β-RPs can reveal dynamical patterns and temporal asymmetries. Although the proposed framework introduces

additional computational cost and requires careful handling of positivity constraints, it can extend the descriptive power of recurrence analysis in some cases.

Future research should focus on several complementary directions. First, extending the framework to joint and cross recurrence analysis may enable the study of coupled dynamical systems under asymmetric similarity measures. Second, reducing the computational burden of β-RP construction is an important objective, particularly when multiple β values must be explored; future work may investigate approximate distance computation, or efficient GPU-based implementations to improve scalability. Third, more principled strategies for handling the strict non-negativity requirement of the β-divergence could be developed, aiming to minimize distortions introduced by shifting, splitting, or regularization while preserving the intrinsic amplitude structure of the signal. Finally, integrating β-RPs into end-to-end learning systems, where β is optimized jointly with model parameters, represents a promising avenue for developing recurrence representations tailored to predictive tasks.

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Appendix

Proof. For $\beta \neq 0, 1$, write

$$d_\beta(x \mid y) = \frac{x^\beta + (\beta-1)y^\beta - \beta x y^{\beta-1}}{\beta(\beta-1)}.$$

Fix $y > 0$ and define

$$\begin{aligned} G_\beta(x) &= \beta(\beta-1)\big[d_\beta(y \mid x) - d_\beta(x \mid y)\big] \\ &= (\beta-2)(x^\beta - y^\beta) + \beta\big(xy^{\beta-1} - yx^{\beta-1}\big). \end{aligned}$$

Then $G_\beta(y) = 0$ and

$$G'_\beta(x) = (\beta-2)\beta x^{\beta-1} + \beta y^{\beta-1} - \beta(\beta-1)y\, x^{\beta-2},$$

hence, $G'_\beta(y) = 0$. Moreover,

$$G''_\beta(x) = \beta(\beta-1)(\beta-2)\, x^{\beta-3}(x-y).$$

Hence, for $x > y$:

- If $\beta \in (-\infty, 0) \cup (1, 2)$, then $G''_\beta(t) < 0$ for all $t > y$, so $G'_\beta(t) < 0$ for all $t \in (y, x]$. Since $G_\beta(y) = 0$,

$$G_\beta(x) - G_\beta(y) = \int_y^x G'_\beta(t)\, dt < 0 \quad \Rightarrow \quad G_\beta(x) < 0.$$

- If $\beta \in (0, 1) \cup (2, \infty)$, then $G''_\beta(t) > 0$ for all $t > y$, so $G'_\beta(t) > 0$ for all $t \in (y, x]$. Since $G_\beta(y) = 0$,

$$G_\beta(x) - G_\beta(y) = \int_y^x G'_\beta(t)\, dt > 0 \quad \Rightarrow \quad G_\beta(x) > 0.$$

Since

$$G_\beta(x) = \beta(\beta-1)\big[d_\beta(y \mid x) - d_\beta(x \mid y)\big],$$

and $\beta(\beta-1) > 0$ for $\beta \in (-\infty, 0) \cup (1, \infty)$, $\beta(\beta-1) < 0$ for $\beta \in (0, 1)$. For $x > y$, we obtain the inequalities stated in (7). The limiting cases $\beta = 1$ and $\beta = 0$ follow by continuity of the divergence with respect to β.

Moreover, for fixed $\beta \neq 2$, we have $G'_\beta(t)$ of constant sign on (y, ∞), hence G_β is strictly monotone on (y, ∞). Since $G_\beta(y) = 0$ and $G_\beta(x)$ keeps a constant sign for $x > y$, it follows that $|G_\beta(x)|$ is strictly increasing in x for $x > y$. Thus, proving the statement for (8).

Finally, we prove (9):

Step 1 (scale reduction). For Lemma 1 we have that $d_\beta(cx \mid cy) = c^\beta d_\beta(x \mid y)$ for any $c > 0$. Hence $\mathcal{R}(\beta)$ is scale-invariant. Set $y = 1$ and write $x = r > 1$.

Step 2 (integral identities). For $\beta \neq 0, 1$, define $p := \beta - 2$ and consider

$$J_+(p) := \int_1^r (t-1)t^p\, dt, \qquad J_-(p) := \int_1^r (r-t)t^p\, dt.$$

A direct evaluation (using $\int t^p\, dt = \frac{t^{p+1}}{p+1}$ and $\int t^{p+1}\, dt = \frac{t^{p+2}}{p+2}$) gives

$$J_+(p) = \left[\frac{t^{p+2}}{p+2} - \frac{t^{p+1}}{p+1}\right]_1^r = \frac{1 + (\beta-1)r^\beta - \beta r^{\beta-1}}{\beta(\beta-1)},$$
$$J_-(p) = \left[\frac{r\, t^{p+1}}{p+1} - \frac{t^{p+2}}{p+2}\right]_1^r = \frac{r^\beta + (\beta-1) - \beta r}{\beta(\beta-1)}.$$

We can obtain the identities

$$d_\beta(1 \mid r) = J_+(p), \qquad d_\beta(r \mid 1) = J_-(p),$$

hence

$$\mathcal{R}(\beta) = \frac{J_+(p)}{J_-(p)}.$$

Step 3 (rewrite the ratio). Let

$$I_0(p) := \int_1^r t^p\, dt, \qquad I_1(p) := \int_1^r t^{p+1}\, dt.$$

Then $J_+(p) = I_1(p) - I_0(p)$ and $J_-(p) = rI_0(p) - I_1(p)$, so

$$\mathcal{R}(\beta) = \frac{I_1 - I_0}{rI_0 - I_1} = g(m(p))\,,$$

$$m(p) := \frac{I_1(p)}{I_0(p)}, \quad g(m) := \frac{m-1}{r-m}.$$

For $t \in [1, r]$, we have $t^{p+1} \geq t^p$ and $t^{p+1} \leq rt^p$. Integrating over $[1, r]$ yields $I_1(p) > I_0(p)$ and $I_1(p) < rI_0(p)$ (strict since $r > 1$), hence $m(p) \in (1, r)$. Therefore

$$g'(m(p)) = \frac{r-1}{(r-m(p))^2} > 0,$$

so it suffices to show that $m(p)$ is strictly increasing in p.

Step 4 (show $m'(p) > 0$).

Recall

$$m(p) = \frac{I_1(p)}{I_0(p)} = \frac{\int_1^r t^{p+1}\, dt}{\int_1^r t^p\, dt} = \frac{\int_1^r t\, t^p\, dt}{\int_1^r t^p\, dt}.$$

Define the positive weight function $w_p(t) = t^p$ on $[1, r]$. Then

$$m(p) = \frac{\int_1^r t\, w_p(t)\, dt}{\int_1^r w_p(t)\, dt},$$

which is the weighted mean of t over $[1, r]$ with weights $w_p(t)$.

If $p_2 > p_1$, then for all $t \in (1, r)$,

$$\frac{w_{p_2}(t)}{w_{p_1}(t)} = t^{p_2 - p_1}$$

is strictly increasing in t. Hence larger values of t receive relatively more weight under w_{p_2} than under w_{p_1}, which shifts the weighted mean to the right. Therefore,

$$m(p_2) > m(p_1),$$

so $m(p)$ is strictly increasing.

□